GCSE Mathematics

Predicted Paper 2 — Higher Tier

Set A

Pearson Edexcel style (1MA1 / 2H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

• Calculator • Ruler graduated in centimetres and millimetres
• Protractor • Pair of compasses • Pen, HB pencil, eraser

Instructions

• Answer **all** 22 questions in the spaces provided.
• You must show all your working.
• Diagrams are NOT accurately drawn unless stated otherwise.
• Calculators may be used.
• If your calculator does not have a π button, take π to be 3.142.

Advice

• Read each question carefully before you start.
• The marks for each question are shown in brackets — use them as a time guide.
• Try to answer every question. Check your answers if you have time at the end.

Answer ALL questions. Write your answers in the spaces provided.

1. In a sale, the price of a coat is reduced by 18%. The sale price of the coat is £73.80. Work out the original price of the coat.

(3 marks)

2. Work out $(8 \times 10^6) \div (4 \times 10^{-2})$. Give your answer in standard form.

(2 marks)

3. A piece of metal has a mass of 540 grams and a volume of 60 cm^3.

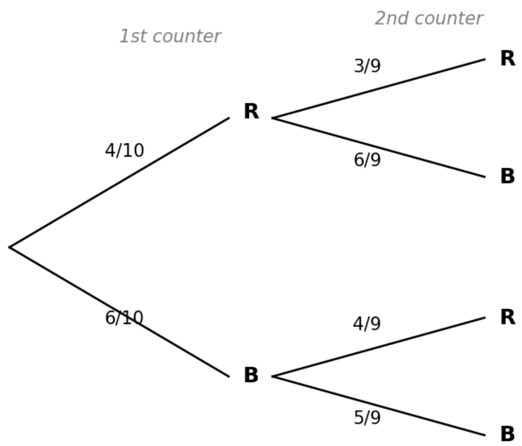
(a) Work out the density of the metal in g/cm^3.

(2 marks)

(b) A second piece of the same metal has a volume of 145 cm^3. Work out the mass of this second piece.

(2 marks)

4. A bag contains 4 red counters and 6 blue counters. Two counters are taken from the bag at random, one after the other, without replacement. The tree diagram below shows the possible outcomes.



(a) Use the tree diagram to find P(both counters are red).

(2 marks)

(b) Work out the probability that the two counters are different colours.

(3 marks)

5. A straight line L passes through the points A(–2, 5) and B(4, –7). Find the equation of line L. Give your answer in the form $y = mx + c$.

(3 marks)

6. Sara invests £4500 in a savings account that pays compound interest at a rate of 2.8% per year. Work out the total amount of interest Sara receives at the end of 3 years. Give your answer to the nearest penny.

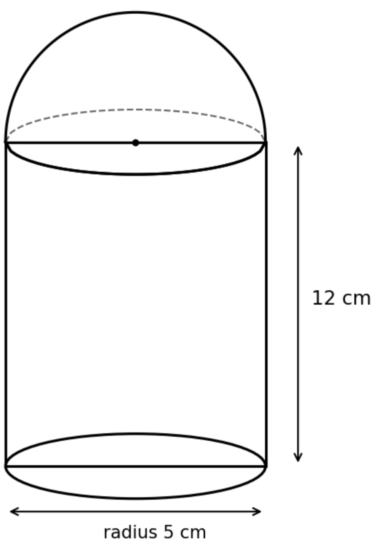
(3 marks)

7. Solve $2x^2 + 7x - 15 = 0$. Show clear algebraic working.

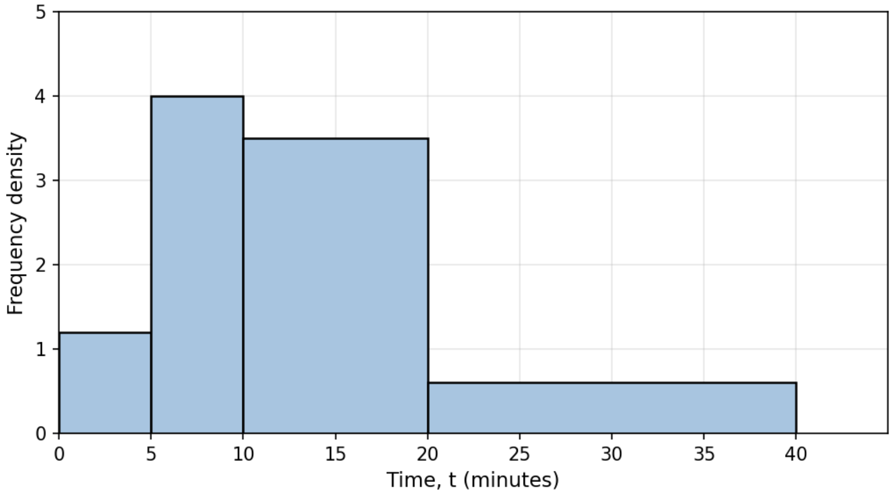
(3 marks)

8. The diagram shows a solid made from a cylinder joined to a hemisphere of the same radius. The cylinder has radius 5 cm and height 12 cm. The hemisphere has radius 5 cm. Work out the total volume of the solid. Give your answer correct to 3 significant figures.

(4 marks)



9. The histogram shows information about the times taken by 80 students to complete a puzzle.



(a) Use the histogram to work out an estimate for the number of students who took longer than 15 minutes.

(3 marks)

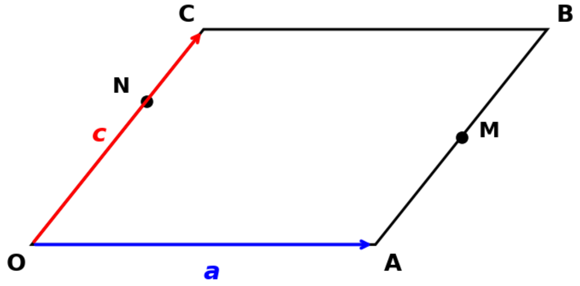
(b) Work out an estimate for the median time taken.

(3 marks)

10. Show that (3 + √5)(2 – √5) can be written in the form a + b√5, where a and b are integers. State the values of a and b.

(3 marks)

11. OABC is a parallelogram. **OA** = **a** and **OC** = **c**. M is the midpoint of AB. N is the point on OC such that ON : NC = 2 : 1.



(a) Express the vector **MN** in terms of **a** and **c**. Give your answer in its simplest form.

(3 marks)

(b) P is a point such that **OP** = λ**a** + **c**. M, N and P lie on a straight line. Find the value of λ.

(3 marks)

12. The ratio of red to blue to green sweets in a jar is 3 : 5 : 7. There are 24 more green sweets than red sweets.

(a) Work out the total number of sweets in the jar.

(3 marks)

(b) A sweet is taken at random and not replaced. Then a second sweet is taken. Work out the probability that both sweets are blue. Give your answer as a fraction in its simplest form.

(3 marks)

13. Simplify fully $(x^2 + 2x - 15) / (x^2 - 9)$.

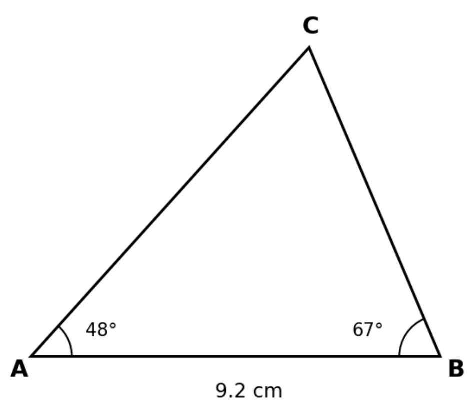
(3 marks)

14. A rectangle has length 12.4 cm and width 7.6 cm, both measured correct to the nearest mm. Work out the upper bound of the perimeter of the rectangle. Show your working clearly.

(3 marks)

15. The diagram shows triangle ABC, where AB = 9.2 cm, angle BAC = 48° and angle ABC = 67°. Work out the length of AC. Give your answer correct to 3 significant figures.

(3 marks)



16. A solution of the equation $x^3 - 4x - 2 = 0$ lies between 2 and 3.

(a) Show that the equation $x^3 - 4x - 2 = 0$ can be rearranged to $x = \sqrt{(4x + 2) / x}$, for $x \neq 0$.

(2 marks)

(b) Starting with $x_0 = 2$, use the iteration formula $x_{n+1} = \sqrt{(4x_n + 2) / x_n}$ to find the values of x_1, x_2 and x_3. Give each value correct to 4 decimal places.

(3 marks)

17. $f(x) = 3x - 4$ and $g(x) = (x + 5) / 2$.

(a) Find $fg(x)$. Give your answer in its simplest form.

(2 marks)

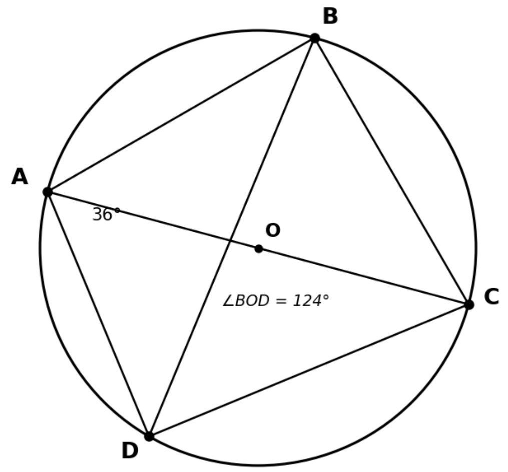
(b) Find the inverse function $f^{-1}(x)$.

(2 marks)

(c) Solve $f(x) = g(x)$.

(3 marks)

18. A, B, C and D are points on the circumference of a circle, centre O. AC is a diameter of the circle. Angle BAC = 36°.



(a) Work out the size of angle ABC. Give a reason for your answer.

(2 marks)

(b) Work out the size of angle ADB. Give a reason for your answer.

(2 marks)

19. y is inversely proportional to the square of x. When x = 4, y = 2.25.

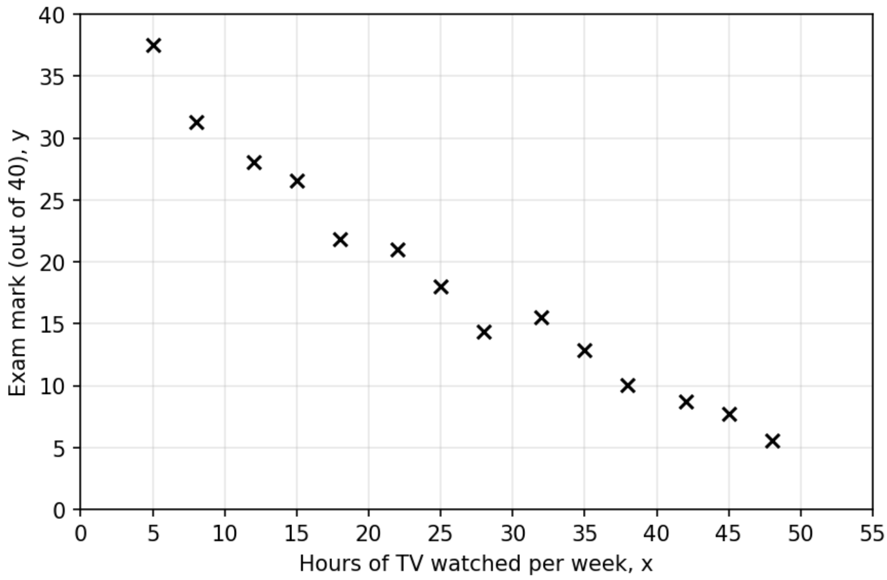
(a) Express y in terms of x.

(2 marks)

(b) Find the positive value of x when y = 9.

(2 marks)

20. The scatter graph shows information about 14 students. It shows the number of hours per week each student watches television (x) and their exam mark out of 40 (y).



(a) Describe the relationship between x and y.

(1 marks)

(b) Another student watches 30 hours of television per week. Use the scatter graph to estimate this student's exam mark.

(2 marks)

(c) A different student watches 80 hours of television per week. Explain why it would not be sensible to use the scatter graph to estimate this student's exam mark.

(1 marks)

21. The curve C has equation $y = x^2 - 6x + 11$.

(a) Write $x^2 - 6x + 11$ in the form $(x - p)^2 + q$, where p and q are integers.

(2 marks)

(b) Hence write down the coordinates of the turning point of C.

(1 marks)

(c) Sketch the curve C on a set of axes, showing clearly the turning point and the y-intercept. (You may sketch in the space below.)

(2 marks)

22. Prove algebraically that the sum of the squares of any two consecutive odd integers is always 2 more than a multiple of 8.

(4 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 2 Set A

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	£90.00	M1: recognises 73.80 represents 82% of original. M1: 73.80 ÷ 0.82. A1: 90.00. **[3]**
2	2×10^8	M1: 8 ÷ 4 = 2 AND $10^6 \div 10^{-2} = 10^8$. A1: 2×10^8 in correct standard form. **[2]**
3(a)	9 g/cm^3	M1: 540 ÷ 60. A1: 9. **[2]**
3(b)	1305 g	M1: 9 × 145. A1: 1305. **[2]**
4(a)	12/90 = 2/15	M1: (4/10) × (3/9). A1: 12/90 oe (= 2/15). **[2]**
4(b)	48/90 = 8/15	M1: identifies P(RB) + P(BR). M1: (4/10)(6/9) + (6/10)(4/9). A1: 48/90 oe. **[3]**
5	y = –2x + 1	M1: gradient = (–7 – 5)/(4 – (–2)) = –2. M1: substitutes a point to find c = 1. A1: y = –2x + 1. **[3]**
6	£388.68	M1: 4500×1.028^3. M1: subtracts 4500. A1: 388.68. **[3]**
7	x = 3/2 or x = –5	M1: (2x – 3)(x + 5). M1: sets factors to zero. A1: both correct. **[3]**
8	1200 cm^3 (3 s.f.)	M1: cylinder = $\pi \times 5^2 \times 12 = 300\pi$. M1: hemisphere = $(2/3)\pi \times 5^3 = 250\pi/3$. M1: adds. A1: 1204... ⇒ 1200. **[4]**
9(a)	29 or 30 (accept 29–30)	M1: frequency in 15–20 = 3.5 × 5 = 17.5. M1: + frequency in 20–40 = 0.6 × 20 = 12. A1: 29.5 (round to 29 or 30). **[3]**
9(b)	Median ≈ 13 minutes	M1: estimate total frequency = 73; half ≈ 36.5. M1: identifies 10–20 class contains median; interpolates 10 + (10.5/35) × 10. A1: 13.0 (accept 12–14). **[3]**
10	a = 1, b = –1	M1: expands four terms: $6 - 3\sqrt{5} + 2\sqrt{5} - 5$. M1: simplifies to $1 - \sqrt{5}$. A1: a = 1, b = –1. **[3]**
11(a)	**MN** = –**a** + (1/6)**c**	M1: **OM** = **a** + (1/2)**c**. M1: **ON** = (2/3)**c**. A1: **MN** = **ON** – **OM** = –**a** + (1/6)**c**. **[3]**
11(b)	λ = –2	M1: **MP** = (λ – 1)**a** + (1/2)**c**. M1: **MP** = k × **MN**, so 1/2 = k/6 ⇒ k = 3 and λ – 1 = –k. A1: λ = –2. **[3]**
12(a)	90 sweets	M1: green – red = 7 – 3 = 4 parts = 24. M1: 1 part = 6. A1: total = 15 × 6 = 90. **[3]**
12(b)	29/267	M1: P(blue first) = 30/90 = 1/3. M1: P(blue second | blue first) = 29/89. A1: (1/3) × (29/89) = 29/267. **[3]**
13	(x + 5) / (x + 3)	M1: numerator (x + 5)(x – 3). M1: denominator (x + 3)(x – 3). A1: (x + 5)/(x + 3). **[3]**
14	40.2 cm	M1: upper bound of length = 12.45. M1: upper bound of width = 7.65. A1: 2(12.45 + 7.65) = 40.2. **[3]**
15	AC ≈ 9.34 cm (3 s.f.)	M1: third angle = 180 – 48 – 67 = 65°. M1: AC / sin 67 = 9.2 / sin 65. A1: 9.34. **[3]**
16(a)	Shown	M1: $x^3 = 4x + 2$, divide by x: $x^2 = (4x + 2)/x$. A1: takes square root. **[2]**
16(b)	$x_1 = 2.2361$, $x_2 = 2.2123$, $x_3 = 2.2145$	M1: $x_1 = \sqrt{5} = 2.2361$. M1: applies iteration twice more. A1: all three values correct to 4 d.p. **[3]**
17(a)	fg(x) = (3x + 7)/2	M1: 3 × (x + 5)/2 – 4. A1: (3x + 7)/2. **[2]**
17(b)	$f^{-1}(x) = (x + 4)/3$	M1: rearranges y = 3x – 4 for x. A1: (x + 4)/3. **[2]**

Q	Answer	Mark scheme
17(c)	$x = 2.6$	M1: $3x – 4 = (x + 5)/2$. M1: $6x – 8 = x + 5$. A1: $x = 13/5 = 2.6$. **[3]**
18(a)	Angle ABC = 90°	B1: 90°. B1: reason "angle in a semicircle". **[2]**
18(b)	Angle ADB = 54°	B1: 54°. B1: reason "angles in the same segment are equal" (angle ACB = 180 – 90 – 36 = 54°). **[2]**
19(a)	$y = 36/x^2$	M1: $y = k/x^2$, $2.25 = k/16$. A1: $k = 36$, $y = 36/x^2$. **[2]**
19(b)	$x = 2$	M1: $9 = 36/x^2 \Rightarrow x^2 = 4$. A1: $x = 2$. **[2]**
20(a)	Negative correlation	B1: identifies negative correlation. **[1]**
20(b)	≈ 14 (accept 12–16)	M1: reads off graph at $x = 30$. A1: value in acceptable range. **[2]**
20(c)	80 is outside the range of the data (extrapolation)	B1: any clear statement that 80 hours is beyond the data range shown. **[1]**
21(a)	$(x – 3)^2 + 2$	M1: $(x – 3)^2 = x^2 – 6x + 9$. A1: + 2, so $p = 3$, $q = 2$. **[2]**
21(b)	(3, 2)	B1ft: turning point from completed square. **[1]**
21(c)	Correct sketch	B1: U-shape with minimum marked at (3, 2). B1: y-intercept at (0, 11). **[2]**
22	Proof shown	M1: lets two consecutive odd integers be $(2n + 1)$ and $(2n + 3)$. M1: expands sum of squares = $8n^2 + 16n + 10$. M1: factorises as $8(n + 1)^2 + 2$. A1: clear conclusion that this is 2 more than a multiple of 8. **[4]**

Worked Solutions — Predicted Paper 2 Set A

Higher Tier · Edexcel 1MA1 style

Q1: Reverse percentage

The sale price represents 100% – 18% = 82% of the original price.

So 82% = £73.80, meaning 1% = 73.80 ÷ 82 = £0.90.

Original price = 100 × 0.90 = **£90.00**.

Check: 90 × 0.82 = 73.80. ✓

Q2: Standard form division

Separate the numbers and the powers of 10:

$(8 \div 4) \times (10^{6} \div 10^{-2}) = 2 \times 10^{6-(-2)} = \mathbf{2 \times 10^{8}}$.

Q3: Density

(a) Density = mass ÷ volume = 540 ÷ 60 = **9 g/cm^3**.

(b) Mass = density × volume = 9 × 145 = **1305 g**.

Q4: Probability tree

(a) P(both red) = P(R then R) = (4/10) × (3/9) = 12/90 = **2/15**.

(b) P(different colours) = P(RB) + P(BR) = (4/10)(6/9) + (6/10)(4/9) = 24/90 + 24/90 = 48/90 = **8/15**.

Q5: Equation of a line

Gradient m = (–7 – 5) / (4 – (–2)) = –12 / 6 = –2.

Using point A(–2, 5): 5 = (–2)(–2) + c, so c = 1.

y = –2x + 1.

Q6: Compound interest

Total amount after 3 years = 4500 × 1.028^3.

= 4500 × 1.086373... = £4888.68 (to nearest penny).

Interest = 4888.68 – 4500 = **£388.68**.

Q7: Quadratic by factorising

Find two numbers that multiply to 2 × (–15) = –30 and sum to 7: +10 and –3.

Split: $2x^2 + 10x - 3x - 15 = 0$.

Group: 2x(x + 5) – 3(x + 5) = (2x – 3)(x + 5) = 0.

So **x = 3/2 or x = –5**.

Q8: Compound solid volume

Cylinder = $\pi r^2 h = \pi \times 25 \times 12 = 300\pi$ cm^3.

Hemisphere = $(2/3)\pi r^3 = (2/3)\pi \times 125 = 250\pi/3$ cm^3.

Total = $300\pi + 250\pi/3 = 1150\pi/3 = 1204.27...$ cm^3.

= **1200 cm^3 (3 s.f.)**.

Q9: Histogram

Frequency = frequency density × class width:

0–5: $1.2 \times 5 = 6$;   5–10: $4.0 \times 5 = 20$;

10–20: $3.5 \times 10 = 35$;   20–40: $0.6 \times 20 = 12$.

(a) Number with $t > 15$: half of the 10–20 bar ($= 5 \times 3.5 = 17.5$) + all of 20–40 bar ($= 12$) = **29.5 (~30 students)**.

(b) Total frequency ≈ 73; half-way at ≈ 36.5. Cumulative reaches 26 at $t = 10$.

Linear interpolation: $10 + (36.5 - 26)/35 \times 10 = 10 + 3.0 =$ **13.0 minutes**.

Q10: Surds

Expand: $(3 + \sqrt{5})(2 - \sqrt{5}) = 6 - 3\sqrt{5} + 2\sqrt{5} - (\sqrt{5})^2$.

$= 6 - 3\sqrt{5} + 2\sqrt{5} - 5 = (6 - 5) + (-3 + 2)\sqrt{5} =$ $\mathbf{1 - \sqrt{5}}$.

So **a = 1, b = –1**.

Q11: Vectors

(a) **OM** = **OA** + (1/2)**AB** = **a** + (1/2)**c**. **ON** = (2/3)**OC** = (2/3)**c**.

MN = **ON** – **OM** = (2/3)**c** – **a** – (1/2)**c** = –**a + (1/6)c**.

(b) **MP** = **OP** – **OM** = (λ – 1)**a** + (1/2)**c**.

For M, N, P collinear: **MP** = k × **MN** for some scalar k.

Coefficients of **c**: $1/2 = k/6$, so $k = 3$.

Coefficients of **a**: $\lambda - 1 = -k = -3$.

So λ **= –2**.

Q12: Ratio + probability

(a) Parts: 3 + 5 + 7 = 15. Green – red = 7 – 3 = 4 parts = 24, so 1 part = 6.

Total = 15 × 6 = **90 sweets**.

(b) Number of blue = 5 × 6 = 30. After one blue is taken: 29 blue, 89 sweets total.

P(both blue) = (30/90) × (29/89) = (1/3) × (29/89) = **29/267**.

Q13: Algebraic fractions

Numerator: $x^2 + 2x - 15 = (x + 5)(x - 3)$.

Denominator: $x^2 - 9 = (x + 3)(x - 3)$.

Cancel $(x - 3)$: **(x + 5) / (x + 3)**.

Q14: Upper bound (perimeter)

Upper bound of length = 12.4 + 0.05 = 12.45 cm.

Upper bound of width = 7.6 + 0.05 = 7.65 cm.

Upper bound of perimeter = 2 × (12.45 + 7.65) = 2 × 20.10 = **40.2 cm**.

Q15: Sine rule

Third angle: angle ACB = 180 – 48 – 67 = 65°.

AC / sin 67 = 9.2 / sin 65.

AC = 9.2 × sin 67 / sin 65 = **9.34 cm (3 s.f.)**.

Q16: Iteration

(a) Start with $x^3 - 4x - 2 = 0$. Rearrange: $x^3 = 4x + 2$.

Divide both sides by x (valid since $x \neq 0$): $x^2 = (4x + 2) / x$.

Take the positive square root: $x = \sqrt{(4x + 2) / x}$. ✓

(b) Starting with $x_0 = 2$:

$x_1 = \sqrt{(4 \times 2 + 2) / 2} = \sqrt{5} =$ **2.2361**.

$x_2 = \sqrt{(4 \times 2.2361 + 2) / 2.2361} = \sqrt{4.8943} =$ **2.2123**.

$x_3 = \sqrt{(4 \times 2.2123 + 2) / 2.2123} = \sqrt{4.9041} =$ **2.2145**.

Q17: Functions

(a) $fg(x) = f((x + 5)/2) = 3 \times (x + 5)/2 - 4 = (3x + 15)/2 - 8/2 =$ **$(3x + 7)/2$**.

(b) Let $y = 3x - 4$. Solve for x: $x = (y + 4)/3$. So **$f^{-1}(x) = (x + 4)/3$**.

(c) $3x - 4 = (x + 5)/2 \Rightarrow 6x - 8 = x + 5 \Rightarrow 5x = 13$. **$x = 2.6$**.

Q18: Circle theorems

(a) AC is a diameter, so angle ABC stands on a semicircle.

By the "angle in a semicircle" theorem, **angle ABC = 90°**.

(b) In triangle ABC: angle ACB = 180 – 90 – 36 = 54°.

Angle ADB and angle ACB are angles in the same segment (both stand on chord AB).

By the "angles in the same segment are equal" theorem, **angle ADB = 54°**.

Q19: Inverse proportion

(a) $y = k / x^2$. When $x = 4$, $y = 2.25$: $2.25 = k / 16$, so $k = 36$.

$y = 36 / x^2$.

(b) $9 = 36 / x^2 \Rightarrow x^2 = 4$. Positive value: **$x = 2$**.

Q20: Scatter graph

(a) As x (TV hours) increases, y (exam mark) decreases. **Negative correlation**.

(b) Reading from the graph at x = 30: y is approximately **14** (acceptable: 12–16).

(c) 80 hours of TV is well outside the data range shown on the graph (5–48 hours).

Extrapolating beyond the data is unreliable, since the trend may not continue.

Q21: Completing the square + sketch

(a) $x^2 - 6x + 11 = (x - 3)^2 - 9 + 11 =$ **$(x - 3)^2 + 2$**. So p = 3, q = 2.

(b) Turning point = **(3, 2)** — a minimum (coefficient of x^2 is positive).

(c) U-shaped parabola, minimum marked at (3, 2), passing through y-intercept (0, 11).

Q22: Algebraic proof

Let two consecutive odd integers be $(2n + 1)$ and $(2n + 3)$, where n is any integer.

Sum of squares = $(2n + 1)^2 + (2n + 3)^2$

$= (4n^2 + 4n + 1) + (4n^2 + 12n + 9)$

$= 8n^2 + 16n + 10$

$= 8(n^2 + 2n + 1) + 2$

$= 8(n + 1)^2 + 2$.

Since $(n + 1)^2$ is always an integer, $8(n + 1)^2$ is always a multiple of 8.

Therefore the sum of squares is always **2 more than a multiple of 8**. ✓